Understanding Sampling Error

Week 7 Tutorial

The Central Problem

What we want: truth about everyone (the population)

What we have: data from a few people (a sample)

How do we bridge the gap *responsibly*?

Example: UBC Food Insecurity

Scenario: AMS seeks permanent foodbank funding from UBC.

- **University's rule:** fund if mean worry ≥ 0.40
- **Our task:** estimate the average fraction of time UBC students worry about their next meal
- **Constraint:** can only survey $n = 2$ students (simple random sample)

Toy population ($N = 7$ students):

Student	1	2	3	4	5	6	7
Worry Y_i	0	0	0	0.5	0.5	1	1

Population mean: $\mu = \dfrac{0+0+0+0.5+0.5+1+1}{7} = \dfrac{3}{7} \approx 0.43$

Truth: students worry 43% of the time $\Rightarrow$ should get funding!

Random Sampling (SRSWOR)

Random sampling means every student has an equal chance of being selected.

How many ways to choose 2 students from 7?

$$\binom{7}{2} = \frac{7 \times 6}{2 \times 1} = 21 \text{ possible samples}$$

Sample mean:

$$\overline{Y} = \frac{1}{2}(Y_{(1)} + Y_{(2)})$$

Key point: In practice, we observe only one $\overline{Y}$, but it could have been any of the 21 outcomes.

Activity: Let's Find Some Samples Together

Task: List a few possible samples and calculate their means.

Examples:

- Sample $\{1, 2\}$: values $(0, 0) \Rightarrow \overline{Y} = \frac{0+0}{2} = 0$

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- Sample $\{1, 3\}$: values $(0, 0) \Rightarrow \overline{Y} = 0$

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- Sample $\{1, 3\}$: values $(0, 0) \Rightarrow \overline{Y} = 0$
- Sample $\{1, 4\}$: values $(0, 0.5) \Rightarrow \overline{Y} = 0.25$

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- Sample $\{1, 3\}$: values $(0, 0) \Rightarrow \overline{Y} = 0$
- Sample $\{1, 4\}$: values $(0, 0.5) \Rightarrow \overline{Y} = 0.25$
- Sample $\{6, 7\}$: values $(1, 1) \Rightarrow \overline{Y} = 1$

Question: Can someone suggest another sample and calculate its mean?

We'll build the full sampling distribution together on the board!

Sampling Distribution of $\overline{Y}$ (all 21 samples)

After enumerating all 21 samples, we get:

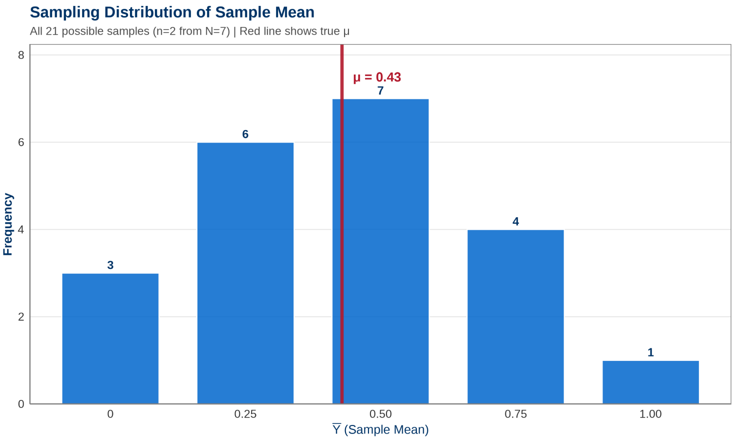
$\overline{Y}$	0	0.25	0.5	0.75	1
Count	3	6	7	4	1
Probability	14.3%	28.6%	33.3%	19.0%	4.8%

Observations:

- Different samples $\Rightarrow$ different $\overline{Y}$ values
- Most common: $\overline{Y} = 0.5$ (appears 7 times)
- Rare extremes: $\overline{Y} = 0$ (3 times) or $\overline{Y} = 1$ (1 time)
- No sample equals $\mu = 3/7 \approx 0.43$ exactly!

Definition: This distribution of all possible $\overline{Y}$ values is the sampling distribution.

Sampling Distribution Visualization



Does your histogram on the board match this?

- Sampling error is unavoidable — every sample differs from μ
- But the distribution is centered near the truth

Key Insight: Unbiasedness

Observation: Every single sample has error — none equals $\mu = 0.43$.

BUT: Let's take the average of all sample means:

$$\begin{aligned}
\text{Average of all } \overline{Y} \text{ values} &= \frac{0 \times 3 + 0.25 \times 6 + 0.5 \times 7 + 0.75 \times 4 + 1 \times 1}{21} \\
&= \frac{0 + 1.5 + 3.5 + 3 + 1}{21} \\
&= \frac{9}{21} = \frac{3}{7} = 0.43 = \mu \quad \checkmark
\end{aligned}$$

Interpretation:

- Random sampling as a procedure is **unbiased**
- On average (over many repetitions), we get the right answer
- This does **NOT** mean any single sample is correct!

Analogy: Like throwing darts — sometimes high, sometimes low, but centered on target.

The Problem: We Observe $\overline{Y} = 0.25$

What happens in practice:

- We conduct the survey once
- We happen to sample students $\{1, 4\}$
- Values: $(0, 0.5) \Rightarrow \overline{Y} = 0.25$

Naive decision rule:

- University funds if $\overline{Y} \geq 0.40$
- We observe $0.25 < 0.40$
- Conclusion: Deny funding

The problem:

- We know the truth: $\mu = 0.43 > 0.40 \Rightarrow$ should fund!
- Our sample led to the wrong decision
- This happened purely by random chance

Key question: How often does random sampling mislead us like this?

Severity: How Often Would We Be Wrong?

Setup:

- Truth: $\mu = 0.43$ (warrants funding)
- Decision threshold: 0.40
- We observed: $\overline{Y} = 0.25 < 0.40$ (would deny funding)

Severity question: If $\mu = 0.43$ (truth warrants funding), how often would random sampling give us $\overline{Y} < 0.40$?

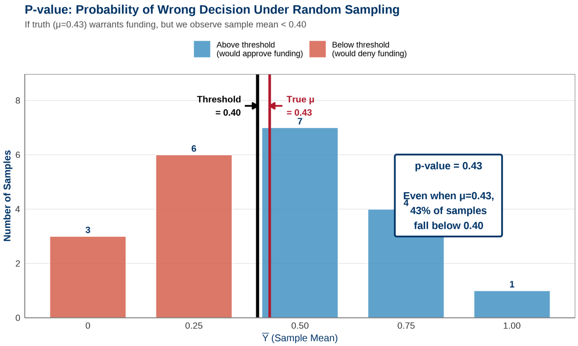
Answer: Look at the sampling distribution:

- Samples with $\overline{Y} < 0.40$: those with $\overline{Y} = 0$ or 0.25
- Count: $3 + 6 = 9$ samples
- Proportion: $p = \frac{9}{21} = \frac{3}{7} \approx 0.43 = 43\%$

Interpretation: Even when truth warrants funding, this procedure would incorrectly deny funding 43% of the time due to random sampling error.

This proportion is called a **p-value**.

P-value Visualization



- Red bars $=$ samples that would lead to wrong decision (9 samples)
- Blue bars $=$ samples that would lead to correct decision (12 samples)
- $p = 9/21 \approx 0.43$ is too high! Not a reliable procedure with $n = 2$.

What P-values Tell Us (and Don't Tell Us)

P-value measures: How likely are we to observe this data (or more extreme) if the null hypothesis were true?

In our example:

- $p = 0.43$ means: "If truth warrants funding, we'd see $\overline{Y} < 0.40$ about 43% of the time by pure chance."
- Lower p-value $\Rightarrow$ less likely to be misled by random error
- Higher p-value $\Rightarrow$ evidence is not "severe"

What p-values do NOT tell us:

- P-values do **not** guard against systematic bias
- P-values do **not** measure the size or importance of an effect
- P-values do **not** tell us the probability that a hypothesis is true

P-values only quantify **random sampling error**, *not bias.*

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Question for you:

In a real UBC survey about food insecurity, which students would be less likely to respond?

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Possible answers:

- Very busy students (multiple jobs, heavy course load)
- Students who don't trust surveys
- Students with **severe** food insecurity (ashamed, don't want to admit)
- International students (language barriers, cultural differences)

What About Sampling Bias?

Question for you:

In a real UBC survey about food insecurity, which students would be less likely to respond?

Possible answers:

- Very busy students (multiple jobs, heavy course load)
- Students who don't trust surveys
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What if one of these groups systematically doesn't respond?

This breaks the "equal probability" assumption of random sampling!

Sampling Bias: Student 7 Never Responds

Scenario: Suppose Student 7 (worry $= 1$) never responds to surveys.

Now the accessible population is $\{0, 0, 0, 0.5, 0.5, 1\}$ (only $N' = 6$ students).

Number of possible samples: $\binom{6}{2} = 15$

New sampling distribution:

$\overline{Y}$	0	0.25	0.5	0.75	1
Count	3	6	4	2	0
Probability	20.0%	40.0%	26.7%	13.3%	0.0%

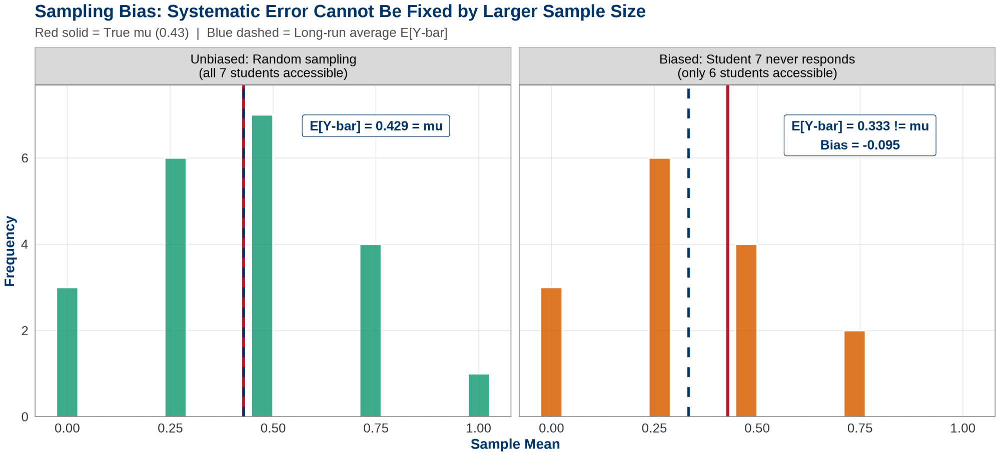
Average of all sample means:

$$\frac{0 \times 3 + 0.25 \times 6 + 0.5 \times 4 + 0.75 \times 2 + 1 \times 0}{15} = \frac{5}{15} = \frac{1}{3} \approx 0.33$$

Bias: $0.33 - 0.43 = -0.10$ (about 22% relative bias)

The procedure is now systematically biased downward!

Bias Comparison Visualization



- **Left:** Random sampling (unbiased) — centered at $\mu = 0.43$
- **Right:** Biased sampling (Student 7 missing) — shifted to 0.33

Why Sampling Bias is Deadly

Key differences:

Random sampling error	**Systematic bias**
• Centered on truth	• Shifted away from truth
• Unbiased on average	• Biased on average
• Reduces with larger n	• Does NOT reduce with larger n
• Quantified by p-values/SE	• P-values do NOT detect bias

Critical lesson:

A large biased sample is worse than a small random sample!

Solution: Design your study carefully to ensure equal probability of inclusion:

- Follow up with non-respondents
- Offer incentives

Key Takeaways

1. **Sampling error is unavoidable**
   - Every sample differs from the population
   - Quantify uncertainty with p-values and confidence intervals
2. **Unbiasedness $\neq$ accuracy**
   - Random sampling is correct *on average* (long run)
   - Any single sample can be way off
3. **P-values only address random error**
   - Low p-value $\Rightarrow$ unlikely due to chance alone
   - P-values do NOT protect against systematic bias
4. **Research design matters more than sample size**
   - Ensure equal probability of selection
   - A biased $n = 10{,}000$ sample $<$ unbiased $n = 100$ sample

Questions?

Appendix: All $\binom{7}{2} = 21$ Samples I

Sample #	Students (i,j)	(Y_i, Y_j)	$\overline{Y}$
1	(1,2)	(0,0)	0.00
2	(1,3)	(0,0)	0.00
3	(1,4)	(0,0.5)	0.25
4	(1,5)	(0,0.5)	0.25
5	(1,6)	(0,1)	0.50
6	(1,7)	(0,1)	0.50
7	(2,3)	(0,0)	0.00
8	(2,4)	(0,0.5)	0.25
9	(2,5)	(0,0.5)	0.25
10	(2,6)	(0,1)	0.50
11	(2,7)	(0,1)	0.50
12	(3,4)	(0,0.5)	0.25
13	(3,5)	(0,0.5)	0.25
14	(3,6)	(0,1)	0.50
15	(3,7)	(0,1)	0.50
16	(4,5)	(0.5,0.5)	0.50
17	(4,6)	(0.5,1)	0.75
18	(4,7)	(0.5,1)	0.75
19	(5,6)	(0.5,1)	0.75
20	(5,7)	(0.5,1)	0.75
21	(6,7)	(1,1)	1.00